

Kirill Burin

kdburin.github.io • github.com/kdburin • linkedin.com/in/kirillburin • kirill.burin@comcast.net • (925) 864-5209

Data Scientist with 1+ years of experience in statistical modeling, machine learning, and feature-level analysis. Skilled in identifying meaningful signals, evaluating feature interactions, and handling noise in real-world datasets to improve model performance.

WORK EXPERIENCE

Turning Green - Data Analyst Intern

09/2025-04/2026

- Analyzed behavioral shifts across 1000+ participants, identifying 3 key trends that were used to refine program curriculum for the 2026 cohorts, using Google Sheets.
- Built visualizations to communicate pre/post changes and key trends to stakeholders using Tableau, Python.
- Conducted parametric and non-parametric statistical tests, including t-tests, ANOVA, Mann-Whitney U, and Wilcoxon signed-rank tests, to quantify program impact and provide evidence for future organizational funding.

Resilience, Inc - Data Analyst Intern

04/2026-Present

- Designed automated Tableau demographic dashboards, cutting stakeholder decision time by 40%.
- Collaborated on data-driven market research to guide strategic organizational projects.
- Engineered 1M+ record data pipelines, reducing monthly manual reporting by 20+ hours.

EDUCATION

Bachelor of Science, Statistical Data Science

06/2024-06/2026

University of California, Davis - Statistics, Linear Algebra, Big Data Analytics, Predictive Modeling, and Deep Learning.

PROJECTS

Predicting Wildfire Incidence in the United States

- Built a county-by-month wildfire prediction dataset by cleaning, filtering, and aggregating U.S. fire records.
- Engineered lagged wildfire features to capture seasonality and past fire activity while avoiding leakage.
- Performed exploratory data analysis and visualizations to identify spatial hotspots and seasonal wildfire patterns.
- Trained and compared logistic regression and XGBoost models to predict notable wildfire occurrence.
- Evaluated model performance with time-aware cross-validation, AUC, accuracy, precision, and F1-score.

Macroeconomic Time Series Analysis

- Built an R-based forecasting pipeline for GDP, exports, and population time series.
- Applied transformations, differencing, and stationarity tests to prepare time series for ARIMA modeling.
- Used variance diagnostics and ACF/PACF plots to identify appropriate ARIMA structures.
- Selected and validated models using AIC/BIC and residual diagnostics, and produced 5-year forecasts.

Swedish Leaf Image Classification

- Developed an image-processing pipeline to classify five Swedish leaf species across 375 samples.
- Extracted and evaluated shape and color features to identify the strongest predictors for distinguishing leaf species.
- Applied EDA, clustering, correlation analysis, ECDFs, and entropy simulations to select discriminative features.
- Trained Random Forest and XGBoost classifiers, achieving 98.67% and 99.20% mean cross-validation accuracy.

Penguin Dataset Analysis

- Used ECDFs and hierarchical clustering to identify natural bins for one or more variables.
- Built species-by-bin contingency tables and converted them into multinomial probability vectors for analysis.
- Compared null and alternative models to test for species-specific patterns versus random bin distribution.
- Simulated observations from multinomial mean vectors to compare entropy distributions across models.
- Evaluated feature importance and species separation, achieving 98% classification accuracy.

SKILLS

Core: Python (Pandas, Scikit-Learn, Statsmodels, Tensorflow, PySpark, PyTorch), R, SQL, Power BI, Tableau.

Machine Learning: Model Evaluation (Precision, Recall, AUC-ROC), Supervised Learning (Logistic Regression, Random Forest, XGBoost, Gradient Boosting), Unsupervised Learning (K-Means, Hierarchical Clustering).

Experimentation & Statistics: A/B Testing, Hypothesis Testing, Multivariate Testing, Probability Theory.

CERTIFICATIONS

Data Analysis with Python - FreeCodeCamp

Machine Learning with Python - Coursera (In progress)